# DISK DEMON
# 2026 Rework

*2026 Rework by NiallEd, October 2026*

During a garage clearout in mid 2024, I discovered my old Commodore 64, 1701 monitor, tape drive and two 1541 disk drives. Along with various floppy disks, tapes and books.

For some reason, just before Christmas 2024, I decided to plug it all together. The monitor worked, the C64 kind of worked but was flashing weird colours all the time. The first disk drive (an Alps) had a ribbon cable hanging out of the back of it and, although it responded to a Load "$",8, it wouldn't read any disks. The second disk drive (a Newtronics) worked perfectly.

That was all it took, I was hooked again - just like back in the 1980s. I spent the next 6 months getting it all working and buying more drives & C64s to improve my repair skills.

The Alps disk drive had this odd looking board installed with a 'Disk Demon' label on one of the ICs. All the other ICs had had their identifying markings removed (I later found out that this had been done to try to prevent copying of the board).

In fact Disk Demon is a floppy disk accelerator for the Commodore 64 and Commodore 128 computers. Although really just a variant of the better known 'Professional DOS', it was sold in the UK as Disk Demon by Datel. It was claimed by the makers to be fastest floppy disk accelerator in the world, able to load a 202 block file in under 3 seconds.

I started to investigate further although initially couldn't find out very much That prompted me to do a lot more digging which was when I discovered Wolfgang Moser's (WoMo) excellent website (unfortunately no longer active) that did a 'deep dive' into Professional DOS. This told me much of what I needed to know and started my journey of discovery.

This project is the result and my attempt to make Disk Demon available in 2026.

Just for fun, I have also included the Professional DOS Demo, translated into English.

Finally, for the sake of clarity, I am neither a hardware nor software engineer. I am just someone with a limited understanding of 1980s electronics who likes to tinker and fix things.

Niall Edwards (NiallEd)

## Acknowledgements

**Wolfgang Moser (WoMo)**
I never met nor had any communication with Wolfgang but without the work he did on Professional DOS back in 2002, this project would never have happened.

**Tim Harris**, **Shareware Plus**.
Always very helpful and provided lots of useful bits & pieces, not least the Professional DOS 1 PCB that Wolfgang had designed.

**FiDo, Lemon 64**.
Also shares an interest in Disk Demon and provided invaluable help and support during the later stages of the project

**The Commodore Community**

## Documentation

The main part of the Disk Demon Instructions is a rework of some original documentation created by Datel back in the mid 1980s. For the most part, I have tried to keep it 'original' and just corrected typos, spelling mistakes and poor grammar that I just could not bring myself to ignore.

I have expanded it a little, adding some basic trouble-shooting and known issues. I have also tested the instructions on the original Disk Demon board and the 2026 rework. Both boards give identical results.

The schematics were drawn using Kicad 10.0.3 for both the original Datel Disk Demon and 2026 Rework.

There is also a PCB layout for the Rework but I have not tried to recreate a PCB for the original Disk Demon.

In addition, all the programming information has been included for the ATF16V8 PLD including a 'dump' of the original 82S123 PROM.

For those who want to make their own Disk Demon 2026 Rework there are Gerber files, a Bill of Materials and step by step assembly tips. It is also available as a Shared Project on PCBWay at https://www.pcbway.com/project/shareproject/Disk_Demon_2026_Rework_C64_Rev3a_214e8d53.html

## Speed tests.

The new board seems fractionally quicker; a regular 202 Block load taking 3.72 seconds on the new board vs 3.85 seconds on the old one. However, that could just be an error in taking the measurements.

Repeating the experiment with a 202 block file and a sector interleave of 1 reduced the load time to around 3.5 seconds.

The option to disable the screen during loading (@DL+) does not work properly using the Disk Demon C64 ROM. However, it does work when using the Professional DOS C64 ROM.

Loading a 202 block file (with a sector interleave of 1) and the screen disabled (using the Professional DOS C64 Kernal ROM) reduces the load time to just over 3 seconds.

That is the closest I have been able to get to the 2.5 seconds load time claimed in the Datel documentation.

## Hardware, Kernals & ROMs

Disk Demon was sold in the UK by Datel and, as far as I can tell, was available for the Commodore 64 and Commodore 128 with a 1541 disk drive connected via the User Port. The Rework is for the Commodore 64. I do not have a kernel ROM for the C128 to test it.

Disk Demon is based on Professional DOS although the schematic is not identical to the one produced by Wolfgang. There are significant differences between the C64 kernal for Disk Demon and C64 kernal for Professional DOS.

The original C64 Disk Demon Kernal ROM was actually a 'dual' ROM. It had a copy of the Disk Demon kernel and a copy of the stock C64 kernal. Datel also provided an adaptor with a toggle switch on a wire to allow each of the two kernals to be easily selected.

I have made BIN files of the original 16k Disk Demon C64 Kernal ROM, an 8k version which just has the Disk Demon Kernal and the 256k EPROM on the Disk Demon board.

**Background**
I believe that Professional DOS was sold in the rest of Europe by mts data GbR. There were options for the C128, 1541C, 1570 & 1571 disk drives. It was also available as an Expansion Port variant.

In the USA, Professional DOS was re-branded as RapiDOS Professional. I don't know much about this version but I think that, like Disk Demon, the main changes were to the C64 kernal.

In the UK, it was sold by Datel as Disk Demon and was available for the C64 with 1541 drive

Each of the kernals has some unique features but they all work with the Disk Demon 2026 Rework board.

Probably the most interesting feature of the Disk Demon 64 kernal is the inclusion of two extra commands; 'FLOAD' and 'FCOPY'. FLOAD allows a file larger than 202 blocks to be loaded (albeit not as quickly) and FCOPY allows a file larger than 202 blocks to be copied.

One aim of the 2026 rework was to use parts that are still widely available in 2026. To that end, the original 82S123 Bipolar PROM has been replaced with an ATF16V8B PLD.

Should anyone want to make their own Disk Demon 2026 Rework for the Commodore 64 and 1541 Disk Drive, I have made the ROMs, Schematics, Bill of Materials and Gerber files available.

Hopefully, someone will be able to adapt it to work with 1571 & 1541-II drives.

I am working on that…

DISK
DEMON

DATEL
ELECTRONICS